# Regulating Rather than Constraining: Adaptive Guidance for Complex Spectral Reconstruction in Pansharpening

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## Abstract

*In remote sensing pansharpening, spectrally mixed regions, where the spectral interactions among adjacent land covers lead to highly inconsistent reconstruction patterns, remain the most challenging areas. Due to the complex spatial distribution and heterogeneous spectral characteristics of ground objects, existing methods relying on rigid architectures and physical constraints struggle to learn generalized reconstruction patterns from limited spectral mixing samples, resulting in unstable generalization. To address this limitation, we propose an architecture-agnostic regularization-guided mechanism that adaptively directs the model to focus on learning reliable reconstruction priors for challenging regions. Specifically, we introduce a simple data-level transformation, MixShuffle, which performs random convex combinations across spatial positions and spectral channels to generate training data with richer spatial structures and stronger spectral mixing. In parallel, we propose a hierarchical attention weighting mechanism, a loss-level gradient reallocation strategy at the sample, channel, and pixel levels, enabling the model to emphasize structurally complex regions. Extensive experiments on multiple benchmark datasets (WV3, GF2, QB) and across various network architectures demonstrate the strong generality and effectiveness of the proposed strategies, achieving state-of-the-art performance when integrated into DANet. Our code is available at `https://github.com/Geo-Tell/DANet`.*

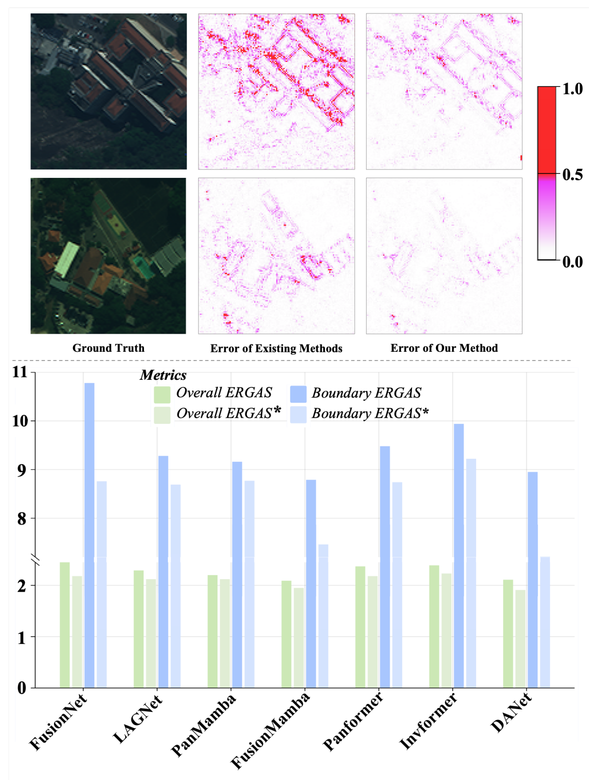


Figure 1. The top row shows the reconstruction error maps of our method and other methods, while the bottom row demonstrates the improvement of our method over different baselines on the WV3 dataset. Here, the suffix '*' indicates that the corresponding baseline incorporates our method.

## 1. Introduction

In remote sensing, high-resolution multispectral (MS) imagery provides both fine spatial details and rich spectral information, substantially improving the accuracy and reliability of various remote sensing tasks [4, 12, 21]. However, due to the physical constraints of sensor design, directly acquiring such data remains highly challenging [24]. This motivates the task of pansharpening, which aims to generate high-resolution MS images by fusing high-resolution panchromatic (PAN) data with low-resolution MS observa-

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tions. Unlike super-resolution for natural images [26, 32], which focuses on producing visually plausible results with relaxed numerical fidelity, pansharpening requires numerically precise reconstruction. Moreover, constrained by spatial resolution, a single pixel in remote sensing imagery often captures a mixed spectral signature from multiple ground objects or materials. This mixing pattern, governed by the diversity of land cover types and their spectral heterogeneity, exhibits high complexity.

Existing pansharpening methods can be broadly categorized into two groups: (1) enhancing feature representation capacity by improving network architectures (e.g., CNNs, Transformers, Mamba) [25, 42, 52], such as adjusting the convolution kernel size to accommodate scale variations across different targets [34]; and (2) introducing modelable physical constraints to strengthen the extraction of specific features [41, 53], for example, leveraging contourlet decomposition or discrete wavelet transform to improve contour-related learning [40, 44]. Although recent methods [15, 34] perform well in reconstructing the spectral characteristics of homogeneous land covers, their performance deteriorates in spectrally mixed regions such as land-cover boundaries and internal textures, as shown in Fig. 1. This suggests that current approaches fail to learn the general reconstruction patterns in spectral mixing regions, and instead rely on superficial, pixel-level approximations.

We argue that the root cause of the aforementioned issues lies in the insufficient attention and learning dedicated to spectrally mixed regions with low pixel proportions during network optimization. Existing methods, which primarily improve performance in specific scenarios through architectural enhancements or the introduction of physical constraints, exhibit inherent limitations: on one hand, relying solely on the inductive biases of the network itself makes it difficult to directly learn generalizable patterns of complex spectral mixing [25, 52]; on the other hand, while predefined physical constraints can strengthen the learning of specific features, they also restrict the model's potential to explore features beyond the imposed priors [41, 44]. In contrast, regularization methods based on data and loss functions introduce inductive assumptions and priors in a more flexible manner [18], allowing for dynamic adjustment of their influence intensity according to training needs, thereby achieving superior spectral reconstruction performance in challenging regions while maintaining model adaptability.

Based on the above analysis, this paper proposes a flexible regularization framework tailored for pansharpening tasks. On the data level, we design a simple data-level transformation, MixShuffle, which performs random convex combinations across training samples and spectral channels to generate augmented data with richer spatial structures and stronger spectral mixing characteristics. On the loss level, we design a hierarchical attention loss (HAL) that employs adaptive gradient reweighting across the sample, channel, and pixel dimensions, guiding the model to focus on structurally complex and challenging regions. In addition, we develop a general dual-scale attention network, DANet, which establishes direct cross-scale spatial–spectral attention interactions, effectively mitigating boundary-blurring issues caused by spatial–spectral misalignment. Extensive experiments demonstrate that the proposed regularization strategies yield substantial performance gains across multiple datasets and diverse baseline models with negligible computational overhead, and they achieve state-of-the-art results when integrated into the DANet architecture (Fig. 1). In summary, the major contributions of this work include:

1. To address the poor generalization in spectrally mixed regions, this paper proposes two general regularization strategies: MixShuffle for data augmentation and HAL for loss optimization. They impose minimal training overhead without affecting inference, while substantially improving the reconstruction of complex spectral structures in a flexible manner.
2. Extensive evaluation on three classic datasets, spanning seven distinct architectures, confirms the method's generality and efficacy. For instance, on the QB dataset, it consistently improved the ERGAS for land-cover boundaries by 4.21%-19.03% across different baselines. Moreover, our dedicated DANet, empowered by this method, achieved a new state-of-the-art performance. On the GF2 dataset, it outperformed the closest competitor by 14.7% in SAM and 5.4% in ERGAS.

## 2. Related work

**Learning-Based Pansharpening Method:** PNN [23] pioneered the introduction of deep learning into pansharpening with a three-layer CNN [20] architecture, achieving significant performance gains and driving continuous innovation in the field. Subsequent research has evolved primarily along two directions: At the network architecture level, PanNet [42] enhances high-frequency information capture using ResNet blocks, MSDCNN [36] employs multi-scale convolutions to adapt to the multi-scale nature of remote sensing images, and ARConv [34] designs learnable convolution kernels to handle varying object sizes. Furthermore, leveraging the long-range modeling capability of Transformers, attention-based models like Panformer [49] and INNformer [52] have emerged. Novel architectures such as FusionMamba [25] utilize Mamba to balance efficiency and performance, while HetSSNet [22] attempts to model spectral-spatial relationships in non-Euclidean space. At the level of embedding physical constraints, GPPNN [41] integrates deep prior regularization and gradient projection to enhance interpretability, FAFNet [40] utilizes discrete wavelet transform to address frequency mismatch between MS and PAN images, and MSDDN [13] deeply explores frequency domain information via Fourier transform; CDFInet [44] introduces contourlet decomposition to capture multi-scale directional fine contour features.

However, these methods generally rely on strong architectural assumptions or physics-based constraints while lacking more flexible regularization mechanisms. Although such rigid priors can enhance performance in specific scenarios, they often fail to maintain effectiveness when transferred across datasets or domains.

**Regularization via Training Data:** Adversarial data augmentation has been extensively adopted in computer vision, where gradient-guided perturbations are introduced to training samples to generate "hard" images that increase model loss [9, 11, 39]. These techniques have been successfully applied to various supervised learning scenarios, including image classification [11, 19] , object detection [5, 9], and segmentation [2, 39]. However, existing methods predominantly focus on universal perturbations rather than task-specific hardness design, which limits their effectiveness in highly structured tasks such as pansharpening, where the key challenge lies in simultaneously preserving fine spatial details and spectral fidelity.

Curriculum learning, inspired by human learning processes, organizes training samples from easy to hard [3, 10, 33]. This paradigm has achieved remarkable success in high-level tasks like image classification [10, 27], object detection [16, 47], and semantic segmentation [35, 46]. Nevertheless, most curriculum-based approaches emphasize difficulty scheduling while neglecting hardness creation, making the training pipeline heavily reliant on the inherent dataset distribution. This limitation is particularly pronounced in low-level vision tasks, where the definition of image "difficulty" remains ambiguous and empirically dependent, constraining broader applications.

Our proposed MixShuffle addresses pansharpening by applying a convex transformation along the spatial-spectral dimensions. This effectively augments the spatial distribution of land cover features while simulating spectral mixing responses.

# 3. Method

**Problem Formulation:** Let $M \in \mathbb{R}^{C\times h\times w}, P \in \mathbb{R}^{1\times H\times W}, F \in \mathbb{R}^{C\times H\times W}$ denote the MS, PAN, and the ground-truth (GT) of the fused image, while $m_i \in \mathbb{R}^{h\times w}, f_i \in \mathbb{R}^{H\times W}(i = 1, 2, \cdots, C)$ represent the (i)-th spectral channel of MS and GT, respectively. The core objective of pansharpening is to find a function $f_{\omega\in\Omega}$ that satisfies $F = f_{\omega\in\Omega}(M, P)$, where $\omega \in \Omega$ represents the learnable parameters. For the training set $\mathcal{D}$, the task can be formulated as finding a weight configuration $\omega^*$ that minimizes the loss function $\mathcal{L}$ as $\omega^* = \arg\min_{\omega\in\Omega} \mathcal{L}(\omega)$. Typically, the loss function takes the form of an expected risk:

$$\mathcal{L} = \frac{1}{|\mathcal{D}|} \sum_{(M_i,P_i,F_i)\in\mathcal{D}} \mathcal{E}(f_{\omega\in\Omega}(M_i, P_i), F_i)\,, \quad (1)$$

where $\mathcal{E}$ is the error function.

**Motivation:** Current pansharpening methods often underperform in spectrally mixed regions (such as landcover boundaries and complex internal textures), leading to blurred edges and spectral distortions [15, 25, 34]. The root cause is that conventional optimization processes treat all image regions equally, causing spectrally mixed features, which are structurally complex yet occupy relatively few pixels, to be effectively overlooked during training. While architectural refinements [25, 52] can introduce certain inductive biases, they struggle to generalize across diverse spectral mixing patterns; meanwhile, methods incorporating fixed physical constraints [41, 44], though enhancing specific feature representation, inherently limit the model's capacity to learn unseen spectral combinations. Therefore, we argue for a shift toward a more flexible and adaptive regularization strategy that guides learning through data and loss functions rather than relying on rigid structural designs. Following this principle, we develop a regularization framework that integrates dynamic data augmentation and hierarchical attention loss, along with a general dual-scale attention network as the backbone model, to systematically address the reconstruction challenges in spectrally mixed regions.

## 3.1. Overview

We propose a flexible regularization framework tailored for the pansharpening task, along with a universal image fusion network, DANet. For the regularization strategies, as illustrated in Fig. 2, we introduce a simple data-level transformation, MixShuffle, which performs random convex combinations across training samples and spectral channels, effectively enriching the diversity of spatial distribution patterns and spectrally mixed regions. In addition, we design a hierarchical attention loss (HAL) that adaptively reallocates gradient weights at the sample, channel, and pixel levels, guiding the model to focus on structurally complex and challenging areas. These regularization strategies are easy to implement, incur only negligible computational cost during training, and introduce no additional overhead during inference. As for DANet, the network establishes direct attention interactions between spatial-spectral features at different scales, effectively mitigating reconstruction blur caused by scale discrepancies and providing a stable structural foundation for the regularization strategies.

## 3.2. MixShuffle

We have designed a convex combination transformation named MixShuffle that operates across spatial and spec-

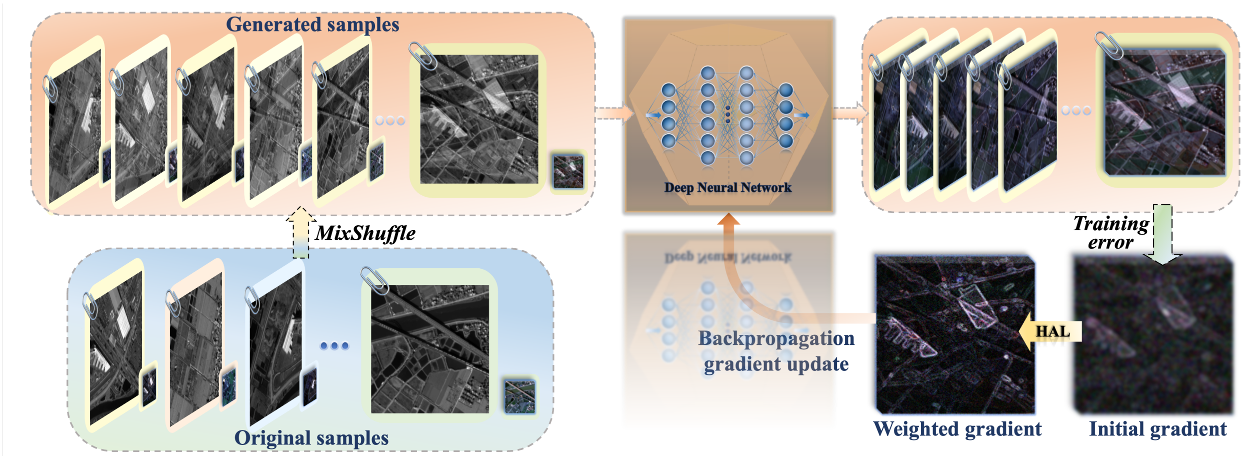


Figure 2. The overview of the proposed method. MixShuffle expands the spatial distribution and spectral mixing responses of ground objects; the Hierarchical Attention Loss (HAL) adaptively emphasizes challenging regions and spectral channels during optimization.

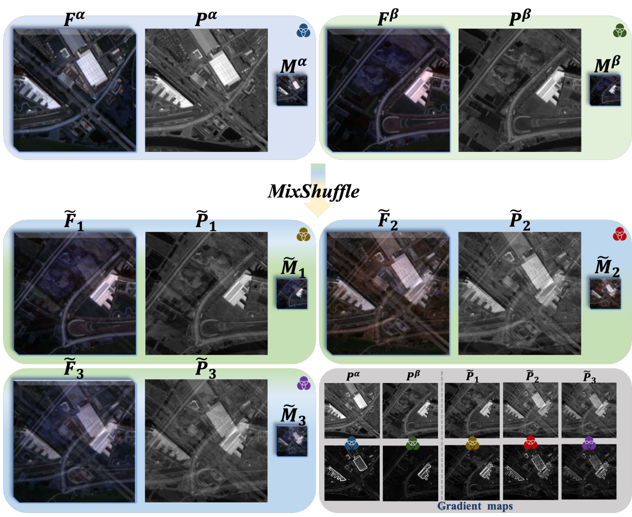


Figure 3. An application example of MixShuffle. Two distinct input samples (first row) are used to generate a series of diverse new samples via MixShuffle.

tral dimensions on the data side. This method significantly expands the spatial distribution of ground objects and the spectrally mixed regions in the training data. As shown in Fig. 3, these synthetically created challenging samples serve as crucial learning materials during network optimization.

For two training samples $(M^\alpha, P^\alpha, F^\alpha)$ and $(M^\beta, P^\beta, F^\beta)$, we first perform sample-level feature mixing to obtain the initial mixed sample $(M^{\alpha\beta}, P^{\alpha\beta}, F^{\alpha\beta})$. This process can be formulated as:

$$\begin{aligned} M^{\alpha\beta} &= \lambda_1 M^\alpha + (1-\lambda_1)M^\beta; \\ P^{\alpha\beta} &= \lambda_1 P^\alpha + (1-\lambda_1)P^\beta; \\ F^{\alpha\beta} &= \lambda_1 F^\alpha + (1-\lambda_1)F^\beta, \end{aligned} \tag{2}$$

where $\lambda_1 \sim \text{Beta}(\theta_1, \theta_1), \theta_1 \in (0, 1)$. Then, through Equation 3, feature mixing is applied to the spectral channels of the initial mixed sample to derive the final fused sample $(\tilde{M}, \tilde{P}, \tilde{F})$, where $S_C$ is the set of all random permutations from 1 to $C$, and $\theta_2 \in (0, 1)$.

$$\begin{aligned} \tilde{m}_i &= \lambda_2 m_i^{\alpha\beta} + (1-\lambda_2) m_{\pi(i)}^{\alpha\beta}, i = 1, 2, \cdots, C; \\ \tilde{f}_i &= \lambda_2 f_i^{\alpha\beta} + (1-\lambda_2) f_{\pi(i)}^{\alpha\beta}, i = 1, 2, \cdots, C; \\ &\text{where } \lambda_2 \sim \text{Beta}(\theta_2, \theta_2); \pi \sim \text{Uniform}(S_C). \end{aligned} \tag{3}$$

Unlike Mixup [48], which only performs linear mixing between samples, MixShuffle not only achieves sample-level feature mixing but also incorporates the core concept of Channel Shuffle [43] to perform random feature mixing across spectral channels.

### 3.3. Hierarchical Attention Loss

In deep learning, the loss function reflects assumptions about the data distribution. To address the challenges of sparse data distribution and complex patterns in spectral mixing regions, we propose a hierarchical attention loss (HAL) that dynamically adjusts gradients for more sufficient optimization in these challenging areas.

Formally, let $\hat{F} \in \mathbb{R}^{C\times H\times W}$ and $F \in \mathbb{R}^{C\times H\times W}$ denote the predicted and ground truth values, respectively, and let $f_{i,j}^c$ and $\hat{f}_{i,j}^c$ represent the values at position $(i, j)$ of the $c$-th channel for the ground truth and prediction. The mathematical expressions for the mean absolute error (MAE) at the pixel, channel, and sample levels are $e_{i,j}^{(c)} = |f_{i,j}^c - \hat{f}_{i,j}^c|$, $e^{(c)} = \frac{1}{HW}\sum_{i,j} e_{i,j}^{(c)}$, and $e = \frac{1}{C}\sum_c e^{(c)}$ respectively. Then, the traditional L1 loss can thus be expressed as:

$$\mathcal{L}_1 = \frac{1}{HWC}\sum_{i,j,c} e_{i,j}^{(c)} = \frac{1}{C}\sum_c e^{(c)} = e. \tag{4}$$

By introducing a polynomial function $\mathcal{W}(x) = x(1+x)^\gamma, \gamma \geq 0$ to the pixel-level, channel-level, and sample-level MAE, we derive three distinct weighted L1 losses:

$$\begin{aligned}\mathcal{L}_{\text{pixel}} &= \frac{1}{HWC}\sum_{i,j,c}\mathcal{W}(e_{i,j}^{(c)});\\ \mathcal{L}_{\text{channel}} &= \frac{1}{C}\sum_{c}\mathcal{W}(e^{(c)});\\ \mathcal{L}_{\text{sample}} &= \mathcal{W}(e).\end{aligned} \tag{5}$$

To enable learning at different hierarchical levels, the HAL combines the three aforementioned weighted losses through a convex combination:

$$\mathcal{L}_{\text{HAL}} = \lambda_{\text{pixel}}\mathcal{L}_{\text{pixel}} + \lambda_{\text{channel}}\mathcal{L}_{\text{channel}} + \lambda_{\text{sample}}\mathcal{L}_{\text{sample}}, \tag{6}$$

where $(\lambda_{pixel}, \lambda_{channel}, \lambda_{sample})$ are the weighting coefficients for the summation.

To intuitively understand the loss behavior, we compute the partial derivatives of $\mathcal{L}_{\text{pixel}}$, $\mathcal{L}_{\text{channel}}$, and $\mathcal{L}_{\text{sample}}$ in Equation 5 with respect to parameter $\Omega$, yielding the results:

$$\begin{aligned}\frac{\partial\mathcal{L}_{\text{pixel}}}{\partial\Omega} &= \frac{1}{HWC}\sum_{i,j,c}\varphi(e_{i,j}^{(c)})\cdot\frac{\partial e_{i,j}^{(c)}}{\partial\Omega};\\ \frac{\partial\mathcal{L}_{\text{channel}}}{\partial\Omega} &= \frac{1}{C}\sum_{c}\varphi(e^{(c)})\cdot\frac{\partial e^{(c)}}{\partial\Omega};\\ \frac{\partial\mathcal{L}_{\text{sample}}}{\partial\Omega} &= \varphi(e)\cdot\frac{\partial e}{\partial\Omega};\\ \text{where } \varphi(x) &= (1+x)^\gamma\left(1+\frac{\gamma x}{1+x}\right).\end{aligned} \tag{7}$$

HAL introduces additional weighting terms $\varphi(\cdot)$ during gradient updates. Two key properties can be observed: (1) In early training stages when errors are large, the gradient magnitude increases significantly, with particularly amplified updates for challenging regions; (2) As errors diminish in later stages, the weighting terms $\varphi(\cdot)$ approach unity, causing the loss to reduce to the standard L1 loss.

### 3.4. DANet Architecture

MS and PAN images exhibit spatial structures and spectral transitions at different scales. However, existing networks often suffer from spatial–spectral misalignment during feature alignment via upsampling or downsampling. To address this issue, we propose a general Dual-scale Attention Network (DANet) that establishes direct attention interactions between spatial–spectral features at different scales, effectively mitigating reconstruction blurring caused by scale discrepancies.

As illustrated in Fig. 4, the DANet consists of convolutional layers, Dual-scale Attention Interaction Modules (DAIMs), and a Dual-scale Attention Fusion Module (DAFM), achieving a concise yet efficient structure.

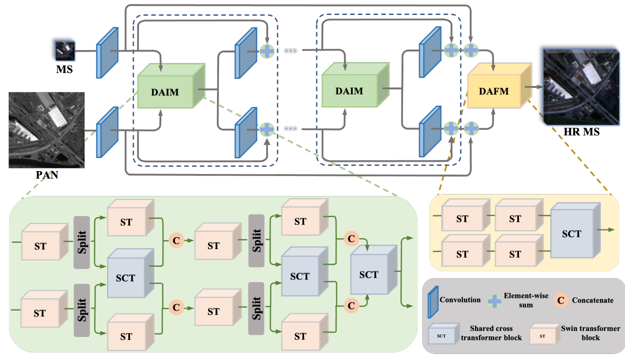


Figure 4. The overall architecture of DANet.

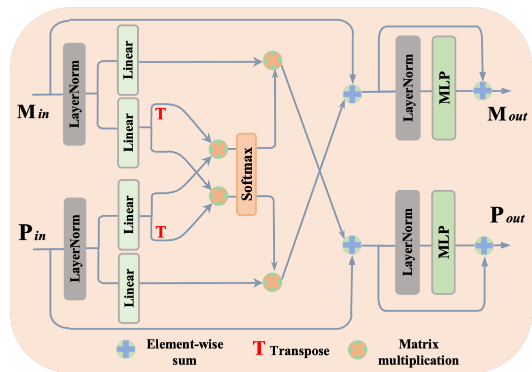


Figure 5. Flowchart of the Shared Cross Transformer (SCT).

The raw input images are first processed by convolutional layers to extract shallow features with higher information density. Subsequently, cascaded DAIMs distill universal spatial-spectral interactive features from these shallow features. Finally, these representative universal features are comprehensively integrated by the DAFM to generate high-definition MS images.

**DAIM:** The DAIM integrates a series of self-attention-based Swin Transformer blocks (STs) and cross-attention-based Shared Cross Transformer blocks (SCTs). Specifically, for input features $X_{\text{in}} \in \mathbb{R}^{M\times C}$ and $Y_{\text{in}} \in \mathbb{R}^{N\times C}$ at distinct scales, independent STs first refine them into attention features $X_1 \in \mathbb{R}^{M\times C}$ and $Y_1 \in \mathbb{R}^{N\times C}$. Inspired by multi-head attention, each token in $X_1$ and $Y_1$ is decoupled into two subtokens, yielding feature components $(X_1^{\text{cross}}, X_1^{\text{self}}) \in \mathbb{R}^{M\times\frac{C}{2}}$ and $(Y_1^{\text{cross}}, Y_1^{\text{self}}) \in \mathbb{R}^{N\times\frac{C}{2}}$. These components then undergo self-interaction via ST and cross-modal interaction via SCT, generating $(X_2^{\text{cross}}, X_2^{\text{self}}) \in \mathbb{R}^{M\times\frac{C}{2}}$ and $(Y_2^{\text{cross}}, Y_2^{\text{self}}) \in \mathbb{R}^{N\times\frac{C}{2}}$. Through token-level concatenation, the interacted components are fused into features $X_2 \in \mathbb{R}^{M\times C}$ and $Y_2 \in \mathbb{R}^{N\times C}$. Similarly, a secondary progressive interaction is applied to $X_2$ and $Y_2$ to derive $X_3$ and $Y_3$, with the final output features $X_{\text{out}} \in \mathbb{R}^{M\times C}$ and $Y_{\text{out}} \in \mathbb{R}^{N\times C}$ generated by a SCT.

Table 1. Quantitative comparison on the WV3 dataset. The best-performing results are in bold, and the second-best are underlined.

| | Reduced Resolution (RR): Avg±std | | | | Full Resolution (FR): Avg±std | | | |
|---|---|---|---|---|---|---|---|---|
| | **SAM↓** | **ERGAS↓** | **Q2n↑** | **SCC↑** | $D_\lambda\downarrow$ | $D_s\downarrow$ | **QNR↑** | **HQNR↑** |
| MTF-GLP-FS [30] | 5.32±1.65 | 4.64±1.44 | 0.818±0.101 | 0.899±0.047 | 0.021±0.008 | 0.063±0.028 | 0.917±0.036 | 0.918±0.035 |
| LRTCFPan [38] | 4.67±1.28 | 4.23±1.28 | 0.836±0.097 | 0.927±0.023 | 0.018±0.007 | 0.053±0.026 | 0.930±0.034 | 0.931±0.031 |
| PanNet [42] | 3.74±0.68 | 2.82±0.71 | 0.862±0.106 | 0.975±0.008 | $\underline{0.017\pm0.007}$ | 0.047±0.021 | 0.936±0.030 | 0.937±0.027 |
| FusionNet [6] | 3.31±0.63 | 2.45±0.59 | 0.896±0.093 | 0.980±0.006 | 0.024±0.009 | 0.036±0.014 | 0.941±0.018 | 0.941±0.020 |
| LAGNet [17] | 3.10±0.50 | 2.29±0.56 | 0.902±0.091 | 0.983±0.006 | 0.037±0.015 | 0.042±0.015 | 0.922±0.027 | 0.923±0.025 |
| Invformer [51] | 3.25±0.64 | 2.39±0.52 | 0.906±0.084 | 0.983±0.005 | 0.055±0.029 | 0.068±0.031 | 0.881±0.053 | 0.882±0.049 |
| DCFNet [37] | 3.03±0.74 | 2.16±0.46 | 0.905±0.088 | 0.986±0.004 | 0.078±0.081 | 0.051±0.034 | 0.875±0.015 | 0.877±0.101 |
| HMPNet [28] | 3.04±0.52 | 2.22±0.49 | 0.913±0.084 | 0.986±0.004 | 0.018±0.007 | 0.053±0.006 | 0.930±0.014 | 0.929±0.011 |
| CANNet [8] | 2.92±0.54 | 2.20±0.47 | 0.914±0.083 | 0.985±0.004 | 0.020±0.009 | 0.029±0.008 | 0.952±0.014 | 0.951±0.010 |
| PanMamba [14] | 2.94±0.54 | 2.20±0.51 | 0.916±0.090 | 0.985±0.006 | 0.020±0.007 | 0.031±0.003 | 0.950±0.065 | 0.954±0.070 |
| FusionMamba [25] | $\underline{2.82\pm0.64}$ | 2.11±0.49 | $\underline{0.920\pm0.091}$ | $\underline{0.989\pm0.005}$ | 0.019±0.008 | $\underline{0.027\pm0.006}$ | 0.955±0.019 | 0.955±0.016 |
| ADWM [34] | 2.89±0.91 | $\underline{1.98\pm0.72}$ | 0.916±0.140 | $\underline{0.989\pm0.006}$ | 0.024±0.020 | 0.033±0.028 | 0.944±0.043 | 0.945±0.019 |
| ARNet [15] | 2.89±0.59 | 2.14±0.53 | 0.910±0.149 | $\underline{0.989\pm0.006}$ | **0.015±0.006** | 0.028±0.007 | $\underline{0.957\pm0.021}$ | $\underline{0.958\pm0.010}$ |
| **Proposed** | **2.69±0.42** | **1.91±0.39** | **0.921±0.131** | **0.991±0.010** | 0.023±0.011 | **0.019±0.009** | **0.958±0.023** | **0.959±0.011** |

**SCT:** Unlike the conventional cross-attention mechanism, the SCT employs a shared query-key matrix strategy to achieve a reduced parameter count. Specifically, for input features $M_{\text{in}} \in \mathbb{R}^{N\times C}$ and $P_{\text{in}} \in \mathbb{R}^{M\times C}$, the standard cross-attention mechanism requires distinct query matrices $(Q_m, Q_p) \in \mathbb{R}^{N\times C}$ and key matrices $(K_m, K_p) \in \mathbb{R}^{M\times C}$ associated with each feature to compute the attention matrices $A_m \in \mathbb{R}^{N\times M}$ and $A_p \in \mathbb{R}^{M\times N}$, respectively. As illustrated in Fig. 5, the SCT utilizes matrices $I_m \in \mathbb{R}^{N\times C}$ and $I_p \in \mathbb{R}^{M\times C}$ in place of the dedicated query and key matrices. Therefore, the calculation formula for the attention matrices become:

$$A_m = \text{Softmax}\left(\frac{I_m I_p^T}{\sqrt{C}}\right), A_p = \text{Softmax}\left(\frac{I_p I_m^T}{\sqrt{C}}\right). \tag{8}$$

# 4. Experiment

This section first introduces the datasets and metrics, then presents the comparative methods, and finally discusses the quantitative and qualitative results.

## 4.1. Datasets and Metrics

**Datasets:** We conducted comprehensive evaluations of the proposed method on three classical yet challenging datasets, including an 8-band dataset WV3 captured by the WorldView-3 sensor and two 4-band datasets QB and GF2 acquired by the QuickBird and GaoFen-2 sensors, respectively. All datasets used in this study are publicly available from open repositories [7]. Additional information about the dataset is provided in the supplementary material.

**Metrics:** Different evaluation metrics were employed according to the spatial resolution of the test sets. Specifically, for the reduced-resolution datasets with ground truth reference data, we employed adopted SAM [45], ERGAS [31], Q2n [29], and SCC [50] to assess reconstruction quality against the ground truth. These ground truth-based metrics provide more stable and direct performance measurements. For the full-resolution datasets lacking ground truth data, we utilized metrics including $D_s$, $D_\lambda$, QNR, and HQNR [1], which assess the similarity between the fused image and the original input images.

## 4.2. Baseline Methods

We selected 14 baseline methods to validate the performance of our approach. These include two traditional methods (MTF-GLP-FS [30] and LRTCFPan [38]) and 12 deep learning-based methods, which can be categorized by their architectural paradigms: CNN-based methods including PanNet [42], FusionNet [6], LAGNet [17], DCFNet [37], HMPNet [28], CANNet [8], ADWM [34], and ARNet [15]; Transformer-based methods including Panformer [49] and Invformer [51]; and Mamba-based methods including PanMamba [14] and FusionMamba [25]. To ensure a fair comparison, all deep learning models were trained and evaluated on the same datasets using their default experimental configurations as recommended in the corresponding publications. For details on the training procedures, please refer to the supplementary material.

## 4.3. Quantitative Results and Analysis

We begin with a comprehensive comparison against state-of-the-art methods on multiple benchmark datasets. The experimental results in Tables 1 and 2 demonstrate the superior performance of our method. On the critical spectral and overall fidelity metrics (SAM and ERGAS), our approach achieves significant improvements over the suboptimal method—4.6% and 14.7% on SAM, and 3.5% and 5.4% on ERGAS for the WV3 and GF2 datasets, respectively. This improvement stems from the regularization techniques MixShuffle and HAL, developed specifically for pansharpening, along with the enhanced scale modeling capability provided by our baseline network DANet.

We further assess the generality of our approach across diverse backbone architectures, including CNN-,

Table 2. Quantitative comparison on the GF2 dataset. The best-performing results are in bold, and the second-best are underlined.

| | Reduced Resolution (RR): Avg±std | | | | Full Resolution (FR): Avg±std | | | |
|---|---|---|---|---|---|---|---|---|
| | SAM↓ | ERGAS↓ | Q2n↑ | SCC↑ | $D_\lambda$↓ | $D_s$↓ | QNR↑ | HQNR↑ |
| MTF-GLP-FS [30] | 1.68±0.35 | 1.62±0.36 | 0.890±0.026 | 0.939±0.016 | 0.035±0.014 | 0.143±0.028 | 0.826±0.030 | 0.823±0.035 |
| LRTCFPan [38] | 1.31±0.28 | 1.30±0.31 | 0.932±0.033 | 0.958±0.013 | 0.033±0.027 | 0.090±0.014 | 0.880±0.026 | 0.881±0.023 |
| PanNet [42] | 1.04±0.19 | 1.02±0.15 | 0.955±0.021 | 0.980±0.003 | 0.021±0.011 | 0.080±0.018 | 0.901±0.024 | 0.901±0.020 |
| FusionNet [6] | 0.98±0.19 | 1.00±0.20 | 0.978±0.005 | 0.978±0.005 | 0.040±0.013 | 0.101±0.013 | 0.863±0.023 | 0.863±0.018 |
| LAGNet [17] | 0.80±0.14 | 0.71±0.11 | 0.979±0.011 | 0.989±0.002 | 0.032±0.013 | 0.079±0.014 | 0.891±0.032 | 0.891±0.020 |
| Invformer [51] | 0.83±0.14 | 0.70±0.11 | 0.977±0.012 | 0.980±0.002 | 0.059±0.026 | 0.110±0.015 | 0.837±0.022 | 0.838±0.024 |
| DCFNet [37] | 0.89±0.16 | 0.81±0.14 | 0.973±0.010 | 0.985±0.002 | 0.023±0.012 | 0.066±0.010 | 0.913±0.017 | 0.912±0.012 |
| HMPNet [28] | 0.80±0.14 | 0.56±0.10 | 0.981±0.030 | 0.993±0.003 | 0.080±0.050 | 0.115±0.012 | 0.814±0.047 | 0.815±0.049 |
| CANNet [8] | 0.72±0.14 | 0.65±0.12 | 0.982±0.007 | 0.991±0.002 | 0.019±0.010 | 0.063±0.009 | 0.919±0.020 | 0.919±0.011 |
| PanMamba [14] | 0.68±0.12 | 0.64±0.10 | 0.982±0.008 | 0.985±0.006 | **0.016±0.008** | 0.045±0.009 | 0.940±0.016 | 0.939±0.010 |
| FusionMamba [25] | 0.71±0.14 | 0.62±0.11 | 0.984±0.007 | 0.995±0.009 | 0.017±0.009 | 0.030±0.007 | 0.952±0.008 | 0.952±0.010 |
| ADWM [34] | 0.68±0.18 | 0.60±0.16 | 0.984±0.016 | 0.996±0.002 | 0.033±0.029 | 0.050±0.033 | 0.920±0.055 | 0.921±0.047 |
| ARNet [15] | 0.70±0.15 | 0.63±0.13 | 0.983±0.010 | 0.991±0.007 | 0.019±0.010 | 0.052±0.001 | 0.930±0.016 | 0.931±0.012 |
| **Proposed** | **0.58±0.13** | **0.53±0.10** | **0.987±0.009** | **0.998±0.002** | 0.024±0.010 | **0.024±0.007** | **0.953±0.018** | **0.953±0.013** |

Table 3. Comparison of baselines on the QB dataset. Methods marked with * are integrated with our approach. Best results are in bold.

| | Reduced Resolution (RR): Avg±std | | | | Full Resolution (FR): Avg±std | | | |
|---|---|---|---|---|---|---|---|---|
| | SAM↓ | ERGAS↓ | Q2n↑ | SCC↑ | $D_\lambda$↓ | $D_s$↓ | QNR↑ | HQNR↑ |
| FusionNet [6] | 4.96±0.84 | 4.19±0.25 | 0.923±0.097 | 0.976±0.010 | 0.059±0.019 | 0.052±0.009 | 0.892±0.031 | 0.892±0.022 |
| FusionNet* | **4.77±0.73** | **4.06±0.22** | **0.925±0.096** | **0.979±0.007** | **0.052±0.016** | **0.049±0.010** | **0.902±0.027** | **0.902±0.026** |
| LAGNet [17] | 4.58±0.77 | 3.87±0.36 | 0.932±0.094 | 0.981±0.009 | 0.084±0.024 | 0.068±0.014 | 0.854±0.021 | 0.854±0.018 |
| LAGNet* | **4.46±0.65** | **3.69±0.42** | **0.934±0.088** | **0.982±0.011** | **0.049±0.034** | **0.056±0.017** | **0.898±0.024** | **0.897±0.021** |
| Panformer [49] | 4.78±0.69 | 3.82±0.25 | 0.926±0.087 | 0.978±0.009 | 0.057±0.036 | 0.061±0.012 | 0.885±0.034 | 0.886±0.029 |
| Panformer* | **4.67±0.58** | **3.68±0.31** | **0.931±0.067** | **0.982±0.011** | **0.036±0.014** | **0.053±0.031** | **0.913±0.072** | **0.913±0.065** |
| Invformer [51] | 4.66±0.78 | 3.70±0.29 | 0.932±0.007 | **0.983±0.007** | 0.174±0.033 | 0.073±0.024 | 0.766±0.038 | 0.766±0.043 |
| Invformer* | **4.44±0.57** | **3.59±0.30** | **0.933±0.010** | **0.983±0.009** | **0.067±0.035** | **0.054±0.029** | **0.883±0.044** | **0.884±0.052** |
| PanMamba [14] | 4.74±0.88 | 4.38±0.60 | 0.923±0.092 | 0.975±0.011 | 0.049±0.013 | **0.044±0.016** | 0.909±0.034 | 0.910±0.027 |
| PanMamba* | **4.66±0.74** | **4.28±0.71** | **0.926±0.077** | **0.978±0.014** | **0.041±0.014** | 0.051±0.017 | **0.910±0.041** | **0.911±0.034** |
| FusionMamba [25] | 4.72±0.96 | 4.61±0.47 | 0.925±0.131 | 0.973±0.012 | 0.040±0.035 | 0.052±0.042 | 0.912±0.063 | 0.911±0.049 |
| FusionMamba* | **4.59±0.77** | **4.53±0.51** | **0.927±0.085** | **0.976±0.019** | **0.038±0.032** | **0.051±0.039** | **0.913±0.059** | **0.914±0.051** |
| ADWM[34] | 4.48±0.87 | 3.67±0.59 | **0.935±0.109** | 0.983±0.010 | **0.038±0.036** | 0.045±0.041 | 0.919±0.062 | 0.918±0.054 |
| ADWM* | **4.44±0.94** | **3.65±0.61** | **0.935±0.098** | **0.984±0.011** | 0.040±0.032 | **0.039±0.045** | **0.923±0.047** | **0.922±0.055** |

Table 4. ERGAS results on WV3 and QB datasets. Asterisked (*) methods are our enhanced baselines; best scores are bolded.

| | WV3 | | QB | |
|---|---|---|---|---|
| | Overall | Boundary | Overall | Boundary |
| FusionNet [6] | 2.45 | 10.78 | 4.19 | 14.03 |
| FusionNet* | **2.18** | **8.76** | **4.06** | **11.36** |
| Improvement (%) | 11.02 | 18.74 | 3.10 | 19.03 |
| PanMamba [14] | 2.20 | 9.16 | 4.38 | 14.95 |
| PanMamba* | **2.12** | **8.77** | **4.28** | **14.17** |
| Improvement (%) | 3.64 | 4.26 | 2.28 | 5.22 |
| FusionMamba [25] | 2.09 | 8.79 | 4.61 | 15.69 |
| FusionMamba* | **1.95** | **7.46** | **4.53** | **15.03** |
| Improvement (%) | 6.70 | 15.13 | 1.74 | 4.21 |
| Invformer [51] | 2.39 | 9.94 | 3.70 | 12.36 |
| Invformer* | **2.23** | **9.22** | **3.59** | **11.32** |
| Improvement (%) | 6.69 | 7.24 | 2.97 | 8.41 |

Transformer-, and Mamba-based networks. As summarized in Table 3, our method consistently delivers significant improvements across all tested architectures. For instance, it enhances the Invformer baseline by 4.7% on the SAM metric and boosts LAGNet by 4.7% on ERGAS. These results demonstrate that the proposed regularization strategies can effectively adapt to the distinct inductive biases of different architectures, thereby enhancing model generalization in a more flexible manner.

Finally, we focus a quantitative analysis on land-cover boundaries—key regions characterized by spectral mixing. As evidenced in Table 4, our method achieves substantial gains in these challenging areas regardless of the backbone architecture or dataset, for example, improving the boundary accuracy of FusionNet [6] by 18.74% on WV3 and 19.03% on QB. This observation is consistent with the motivation behind our MixShuffle and HAL regularization techniques tailored for pansharpening. Additional results and benchmarks are provided in the supplementary material.

### 4.4. Qualitative Results and Analysis

In addition to quantitative results, we conducted a corresponding qualitative evaluation, as shown in Fig. 6 and 7. By examining the error maps between the pansharpening results and the ground truth in Fig. 6, we observe that existing

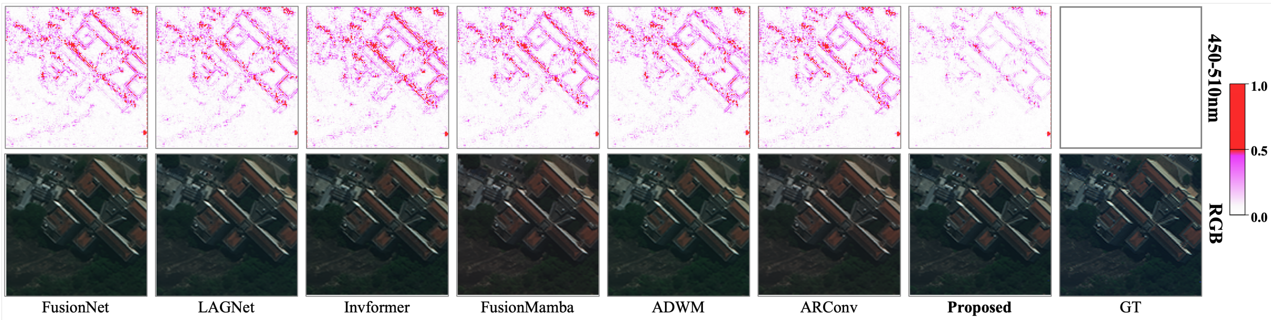


Figure 6. Qualitative comparison of different methods on WV3 dataset. Top row: reconstruction error maps; Bottom row: RGB outputs.

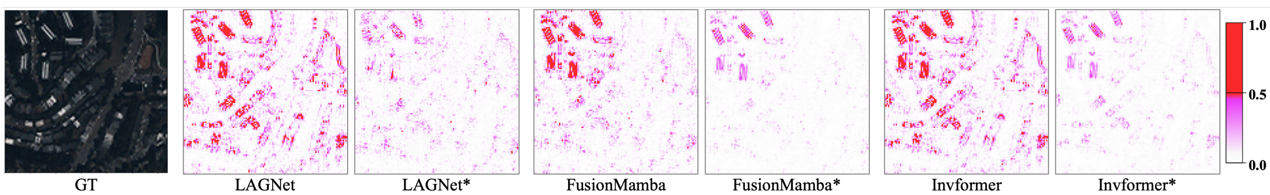


Figure 7. Reconstruction error comparison of different methods on the QB dataset. The suffix '*' denotes integration with our method.

methods exhibit the most significant errors along land-cover boundaries and within textured areas, where complex spectral mixing responses substantially increase the difficulty of reconstruction. In contrast, our method achieves the lowest reconstruction errors among all compared approaches, particularly excelling in challenging transition regions such as those between building rooftops and the ground, or between rooftops and surrounding walls by effectively capturing the intrinsic spectral mixing patterns.

Furthermore, the visual assessment in Fig. 7 demonstrates that integrating our method consistently enhances the reconstruction quality across various baseline architectures. This improvement is particularly evident in object boundaries and textured areas, such as the roof edges and internal junctions of the buildings in the figure. These regions exhibit complex spectral mixing patterns owing to diverse materials or designs, the reconstruction fidelity of which is markedly improved with our approach. More visual results are provided in the supplementary material.

### 4.5. Ablation Study

To validate the effectiveness of the proposed components, we conduct systematic ablation studies based on our DANet across three datasets. As shown in Table 5, we evaluate the individual and combined contributions of the MixShuffle and the HAL. The results demonstrate that both components independently contribute to performance gains. Specifically, introducing either the MixShuffle or the HAL alone leads to consistent improvements across all evaluation metrics on all three datasets. More importantly, their joint utilization achieves the best performance. For instance, on the WV3 dataset, our full model equipped with both components significantly reduces the ERGAS score from 2.07 to 1.91 compared to the baseline.

Table 5. Ablation experiment on WV3, GF2, and QB datasets.

| | MixShuffle | HAL | SAM↓ | ERGAS↓ | Q2n↑ | SCC↑ |
|---|---|---|---|---|---|---|
| WV3 | | | 2.85 | 2.07 | 0.912 | 0.988 |
| | ✓ | | 2.74 | 1.96 | 0.918 | 0.990 |
| | | ✓ | 2.78 | 1.99 | 0.916 | 0.989 |
| | ✓ | ✓ | **2.69** | **1.91** | **0.921** | **0.991** |
| GF2 | | | 0.62 | 0.59 | 0.983 | 0.994 |
| | ✓ | | 0.59 | 0.55 | 0.986 | 0.997 |
| | | ✓ | 0.58 | 0.57 | 0.985 | **0.998** |
| | ✓ | ✓ | **0.58** | **0.53** | **0.987** | **0.998** |
| QB | | | 4.60 | 4.10 | 0.926 | 0.979 |
| | ✓ | | 4.43 | 3.70 | 0.934 | 0.982 |
| | | ✓ | 4.45 | 3.72 | 0.934 | 0.981 |
| | ✓ | ✓ | **4.38** | **3.67** | **0.935** | **0.984** |

## 5. Conclusion

In this paper, we address the generalization challenge in spectrally mixed regions for pansharpening by proposing a flexible regularization framework. On the data side, MixShuffle extends spatial distribution of ground objects to simulate spectral mixing responses, while on the loss side, HAL adaptively emphasizes difficult regions and spectral channels during optimization. Their joint effect introduces dynamic regularization that significantly improves generalization and robustness. Extensive experiments across diverse datasets and architectures validate the effectiveness and universality of our approach.

## References


[1] Alberto Arienzo, Gemine Vivone, Andrea Garzelli, Luciano Alparone, and Jocelyn Chanussot. Full-resolution quality assessment of pansharpening: Theoretical and hands-on approaches. *IEEE Geoscience and Remote Sensing Magazine*, 10(3):168–201, 2022. 6

[2] Anurag Arnab, Ondrej Miksik, and Philip H.S. Torr. On the robustness of semantic segmentation models to adversarial attacks. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018. 3

[3] Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, 2009. 3

[4] Francesca Bovolo, Lorenzo Bruzzone, Luca Capobianco, Andrea Garzelli, Silvia Marchesi, and Filippo Nencini. Analysis of the effects of pansharpening in change detection on vhr images. *IEEE Geoscience and Remote Sensing Letters*, 7(1):53–57, 2010. 1

[5] Shang-Tse Chen, Cory Cornelius, Jason Martin, and Duen Horng (Polo) Chau. Shapeshifter: Robust physical adversarial attack on faster r-cnn object detector. In *Machine Learning and Knowledge Discovery in Databases*, pages 52–68, Cham, 2019. Springer International Publishing. 3

[6] Liang-Jian Deng, Gemine Vivone, Cheng Jin, and Jocelyn Chanussot. Detail injection-based deep convolutional neural networks for pansharpening. *IEEE Transactions on Geoscience and Remote Sensing*, page 6995–7010, 2021. 6, 7

[7] Liang-jian Deng, Gemine Vivone, Mercedes E. Paoletti, Giuseppe Scarpa, Jiang He, Yongjun Zhang, Jocelyn Chanussot, and Antonio Plaza. Machine learning in pansharpening: A benchmark, from shallow to deep networks. *IEEE Geoscience and Remote Sensing Magazine*, 10(3): 279–315, 2022. 6

[8] Yule Duan, Xiao Wu, Haoyu Deng, and Liang-Jian Deng. Content-adaptive non-local convolution for remote sensing pansharpening. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27738–27747, 2024. 6, 7

[9] Kevin Eykholt, Ivan Evtimov, Earlence Fernandes, Bo Li, Amir Rahmati, Florian Tramèr, Atul Prakash, Tadayoshi Kohno, and Dawn Song. Physical adversarial examples for object detectors. *CoRR*, abs/1807.07769, 2018. 3

[10] Chen Gong, Dacheng Tao, Stephen J. Maybank, Wei Liu, Guoliang Kang, and Jie Yang. Multi-modal curriculum learning for semi-supervised image classification. *IEEE Transactions on Image Processing*, page 3249–3260, 2016. 3

[11] Ian Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *Cornell University - arXiv,Cornell University - arXiv*, 2014. 3

[12] Zhu Han, Shuyi Xu, Lianru Gao, Zhi Li, and Bing Zhang. Gretnet: Gaussian retentive network for hyperspectral image classification. *IEEE Geoscience and Remote Sensing Letters*, 21:1–5, 2024. 1

[13] Xuanhua He, Keyu Yan, Jie Zhang, Rui Li, Chengjun Xie, Man Zhou, and Danfeng Hong. Multiscale dual-domain guidance network for pan-sharpening. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–13, 2023. 3

[14] Xuanhua He, Ke Cao, Jie Zhang, Keyu Yan, Yingying Wang, Rui Li, Chengjun Xie, Danfeng Hong, and Man Zhou. Panmamba: Effective pan-sharpening with state space model. *Information Fusion*, 115:102779, 2025. 6, 7

[15] Jie Huang, Haorui Chen, Jiaxuan Ren, Siran Peng, and Liangjian Deng. A general adaptive dual-level weighting mechanism for remote sensing pansharpening. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2, 3, 6, 7

[16] SouYoung Jin, Aruni RoyChowdhury, Huaizu Jiang, Ashish Singh, Aditya Prasad, Deep Chakraborty, and Erik Learned-Miller. Unsupervised hard example mining from videos for improved object detection. In *Computer Vision – ECCV 2018*, pages 316–333, Cham, 2018. Springer International Publishing. 3

[17] Zi-Rong Jin, Tian-Jing Zhang, Tai-Xiang Jiang, Gemine Vivone, and Liang-Jian Deng. Lagconv: Local-context adaptive convolution kernels with global harmonic bias for pansharpening. *Proceedings of the AAAI Conference on Artificial Intelligence*, page 1113–1121, 2022. 6, 7

[18] Jan Kukačka, Vladimir Golkov, and Daniel Cremers. Regularization for deep learning: A taxonomy. *Cornell University - arXiv,Cornell University - arXiv*, 2017. 2

[19] Alexey Kurakin, Ian J Goodfellow, and Samy Bengio. Adversarial examples in the physical world. In *Artificial intelligence safety and security*, pages 99–112. Chapman and Hall/CRC, 2018. 3

[20] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, page 2278–2324, 1998. 2

[21] Junmin Liu, Shijie Li, Changsheng Zhou, Xiangyong Cao, Yong Gao, and Bo Wang. Sraf-net: A scene-relevant anchor-free object detection network in remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, page 1–14, 2022. 1

[22] Mengting Ma, Yizhen Jiang, Mengjiao Zhao, Jiaxin Li, and Wei Zhang. Hetssnet: Spatial-spectral heterogeneous graph learning network for panchromatic and multispectral images fusion. *arXiv preprint arXiv:2502.04623*, 2025. 2

[23] Giuseppe Masi, Davide Cozzolino, Luisa Verdoliva, and Giuseppe Scarpa. Pansharpening by convolutional neural networks. *Remote Sensing*, page 594, 2016. 2

[24] Xiangchao Meng, Huanfeng Shen, Huifang Li, Liangpei Zhang, and Randi Fu. Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges. *Information Fusion*, page 102–113, 2019. 1

[25] Siran Peng, Xiangyu Zhu, Haoyu Deng, Liang-Jian Deng, and Zhen Lei. Fusionmamba: Efficient remote sensing image fusion with state space model. *IEEE Transactions on Geoscience and Remote Sensing*, page 1–1, 2024. 2, 3, 6, 7

[26] Xibin Song, Yuchao Dai, Dingfu Zhou, Liu Liu, Wei Li, Hongdong Li, and Ruigang Yang. Channel attention based iterative residual learning for depth map super-resolution. In *Proceedings of the ieee/cvf conference on computer vision and pattern recognition*, pages 5631–5640, 2020. 2

[27] Yuxing Tang, Xiaosong Wang, Adam P. Harrison, Le Lu, Jing Xiao, and Ronald M. Summers. Attention-guided curriculum learning for weakly supervised classification and localization of thoracic diseases on chest radiographs. In *Machine Learning in Medical Imaging*, pages 249–258, Cham, 2018. Springer International Publishing. 3

[28] Xin Tian, Kun Li, Wei Zhang, Zhongyuan Wang, and Jiayi Ma. Interpretable model-driven deep network for hyperspectral, multispectral, and panchromatic image fusion. *IEEE Transactions on Neural Networks and Learning Systems*, 35 (10):14382–14395, 2024. 6, 7

[29] Gemine Vivone, Luciano Alparone, Jocelyn Chanussot, Mauro Dalla Mura, Andrea Garzelli, Giorgio A. Licciardi, Rocco Restaino, and Lucien Wald. A critical comparison among pansharpening algorithms. *IEEE Transactions on Geoscience and Remote Sensing*, 53(5):2565–2586, 2015. 6

[30] Gemine Vivone, Rocco Restaino, and Jocelyn Chanussot. Full scale regression-based injection coefficients for panchromatic sharpening. *IEEE Transactions on Image Processing*, page 3418–3431, 2018. 6, 7

[31] Lucien Wald. *Data fusion: definitions and architectures: fusion of images of different spatial resolutions*. Presses des MINES, 2002. 6

[32] Tengfei Wang, Jiaxin Xie, Wenxiu Sun, Qiong Yan, and Qifeng Chen. Dual-camera super-resolution with aligned attention modules. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 2

[33] Xin Wang, Yudong Chen, and Wenwu Zhu. A survey on curriculum learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, page 1–1, 2021. 3

[34] Xueyang Wang, Zhixin Zheng, Jiandong Shao, Yule Duan, and Liang-Jian Deng. Adaptive rectangular convolution for remote sensing pansharpening. *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17872–17881, 2025. 2, 3, 6, 7

[35] Yunchao Wei, Xiaodan Liang, Yunpeng Chen, Xiaohui Shen, Ming-Ming Cheng, Jiashi Feng, Yao Zhao, and Shuicheng Yan. Stc: A simple to complex framework for weakly-supervised semantic segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, page 2314–2320, 2017. 3

[36] Yancong Wei, Qiangqiang Yuan, Xiangchao Meng, Huanfeng Shen, Liangpei Zhang, and Michael Ng. Multi-scale-and-depth convolutional neural network for remote sensed imagery pan-sharpening. In *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 2017. 2

[37] Xiao Wu, Ting-Zhu Huang, Liang-Jian Deng, and Tian-Jing Zhang. Dynamic cross feature fusion for remote sensing pansharpening. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 6, 7

[38] Zhong-Cheng Wu, Ting-Zhu Huang, Liang-Jian Deng, Jie Huang, Jocelyn Chanussot, and Gemine Vivone. Lrtcfpan: Low-rank tensor completion based framework for pansharpening. *IEEE Transactions on Image Processing*, 32:1640–1655, 2023. 6, 7

[39] Cihang Xie, Jianyu Wang, Zhishuai Zhang, Yuyin Zhou, Lingxi Xie, and Alan Yuille. Adversarial examples for semantic segmentation and object detection. In *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017. 3

[40] Yinghui Xing, Yan Zhang, Houjun He, Xiuwei Zhang, and Yanning Zhang. Pansharpening via frequency-aware fusion network with explicit similarity constraints. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–14, 2023. 2

[41] Shuang Xu, Jiangshe Zhang, Zixiang Zhao, Kai Sun, Junmin Liu, and Chunxia Zhang. Deep gradient projection networks for pan-sharpening. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2, 3

[42] Junfeng Yang, Xueyang Fu, Yuwen Hu, Yue Huang, Xinghao Ding, and John Paisley. Pannet: A deep network architecture for pan-sharpening. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 1753–1761, 2017. 2, 6, 7

[43] Huaxiu Yao, Long-Kai Huang, Linjun Zhang, Ying Wei, Li Tian, James Y. Zou, Junzhou Huang, and Zhenhui Jessie Li. Improving generalization in meta-learning via task augmentation. In *International Conference on Machine Learning*, 2020. 4

[44] Xiaoyu Yi, Hao Zhu, Pute Guo, Biao Hou, Bo Ren, Xiaotong Li, Xiaoteng Wang, and Licheng Jiao. Contour-aware dynamic low-high frequency integration for pan-sharpening. *IEEE Transactions on Geoscience and Remote Sensing*, 63: 1–13, 2025. 2, 3

[45] Roberta H. Yuhas, Alexander F. H. Goetz, and Joe W. Boardman. Discrimination among semi-arid landscape endmembers using the spectral angle mapper (sam) algorithm. In *JPL, Summaries of the Third Annual JPL Airborne Geoscience Workshop. Volume 1: AVIRIS Workshop*, 1992. 6

[46] Dingwen Zhang, Le Yang, Deyu Meng, Dong Xu, and Junwei Han. Spftn: A self-paced fine-tuning network for segmenting objects in weakly labelled videos. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. 3

[47] Dingwen Zhang, Junwei Han, Long Zhao, and Deyu Meng. Leveraging prior-knowledge for weakly supervised object detection under a collaborative self-paced curriculum learning framework. *International Journal of Computer Vision*, page 363–380, 2019. 3

[48] Hongyi Zhang, Moustapha Cisse, YannN. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. *arXiv preprint arXiv:1710.09412*, 2017. 4

[49] Huanyu Zhou, Qingjie Liu, and Yunhong Wang. Panformer: A transformer based model for pan-sharpening. In *2022 IEEE International Conference on Multimedia and Expo (ICME)*, pages 1–6, 2022. 2, 6, 7

[50] J. Zhou, D. L. Civco, and J. A. Silander. A wavelet transform method to merge landsat tm and spot panchromatic data. *International Journal of Remote Sensing*, page 743–757, 1998. 6

[51] Man Zhou, Xueyang Fu, Jie Huang, Feng Zhao, Aiping Liu, and Rujing Wang. Effective pan-sharpening with transformer and invertible neural network. *IEEE Transactions on Geoscience and Remote Sensing*, page 1–15, 2022. 6, 7

[52] Man Zhou, Jie Huang, Yanchi Fang, Xueyang Fu, and Aiping Liu. Pan-sharpening with customized transformer and

invertible neural network. *Proceedings of the AAAI Conference on Artificial Intelligence*, page 3553–3561, 2022. 2, 3

[53] Man Zhou, Keyu Yan, Jie Huang, Zihe Yang, Xueyang Fu, and Feng Zhao. Mutual information-driven pan-sharpening. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1788–1798, 2022. 2